

Does Online Gravity Estimation Matter? Revisiting a Silent Design Split in LiDAR-Inertial Odometry

Jie Xu Ziyi Jin Kangjin Yu Hongjun Huang Tongxing Jin Hongkun Luo Zhongpu Xia

Anyverse Dynamics

Abstract—LiDAR-inertial odometry (LIO) systems differ in whether they continue estimating gravity after initialization. We compare four gravity-bias state configurations in each of FAST-LIO2 and LIO-SAM, then separately test a gravity-direction factor. Across 12 dataset sequences evaluated with FAST-LIO2, fixing gravity under continuous LiDAR correction produces mean paired changes in vertical and 3D position errors with 90% confidence intervals within $\pm 2\%$. Tests on 4 sequences with LIO-SAM likewise show no consistent benefit from online gravity. Multi-second LiDAR outages, unlike reduced range or field of view, reveal trajectory-dependent costs of fixing gravity. A history-matched 23D-to-21D switch places the repeatable 3D error increase after LiDAR updates resume. Under 5-s outages, a direction factor from the same IMU used for preintegration improves accuracy on Hall05 but worsens both errors with online gravity on TUHH. Dynamic-start tests also show fixed-bias failures at particular starting phases. We recommend keeping gravity and accelerometer bias online for robustness; use a direction factor only after verifying vertical and 3D accuracy gains under the intended operating conditions.

Code, evidence, and video:

<https://github.com/jiejie567/llo-gravity-ablation>

I. INTRODUCTION

LiDAR–inertial odometry (LIO) combines inertial propagation with corrections from scan registration. This propagation requires gravity and accelerometer bias, but systems differ in whether these quantities remain estimated states after initialization. FAST-LIO2 and Point-LIO retain gravity direction on S^2 online [1], [2], whereas LIO-SAM fixes it [3]. Lightning-LM goes further, reducing its filter from 23D to 12D by removing online estimation of accelerometer bias $\mathbf{b}_a$, gravity $\mathbf{g}$, and the extrinsics [4]. These designs raise a practical question: which states need to remain online, and under what operating conditions?

Comparing published trajectory errors cannot answer this question. The systems also differ in registration, maps, trajectory models, and parameter settings. Even within one system, a changing gravity estimate does not establish that estimating it improves pose. Under weak excitation, changes in gravity, attitude, and accelerometer bias can explain similar inertial residuals. Gravity may then wander while LiDAR corrections keep the trajectory accurate. Removing gravity could suppress this behavior, but could also remove an adjustment needed after poor initialization or a long interruption of LiDAR correction.

Gravity adds only two tangent coordinates, so removing it offers little computational incentive if point-cloud matching dominates runtime. The stronger reason to fix gravity would

Corresponding author: Zhongpu Xia. Contact (Jie Xu): jeff_xu_0503@foxmail.com.

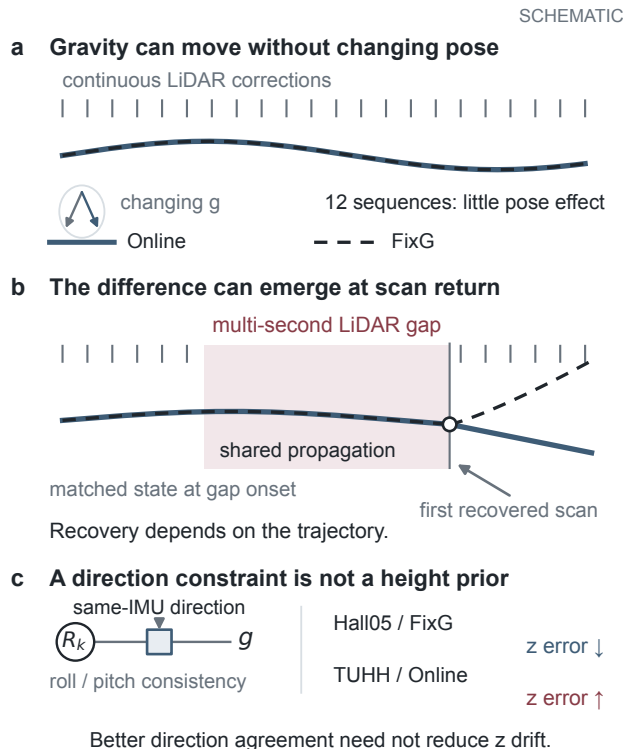


Fig. 1. Study logic (schematic, not measured paths). **a**, Continuous LiDAR correction can mask gravity-state motion. **b**, Matched states share propagation; scan return can reveal different recovery. **c**, A direction factor from the same IMU constrains attitude, not height: the tested dropout cases show opposite vertical-error responses.

be fewer weakly observable states without losing accuracy or recovery. Testing that argument requires conditions beyond nominal operation, where frequent scans may conceal the consequences of either choice.

We test whether online gravity estimation remains necessary for accurate pose estimation when LiDAR updates are frequently accepted. We then weaken the available geometry or remove corrections altogether to determine whether these conditions have the same effect. Dynamic startup tests a different source of error: motion during the shared acceleration-mean initialization. Keeping $\mathbf{g}$ and $\mathbf{b}_a$ online permits subsequent adjustment, although their individual estimates need not become physically correct. The experiments distinguish improved trajectory estimation from accurate identification of these two coupled quantities.

Adding a gravity-direction factor is a separate design

choice. Fixing gravity removes a state, whereas the factor adds a residual that constrains attitude relative to an estimated down direction. It does not measure height. Any vertical-error reduction must arise indirectly through attitude, inertial propagation, or scan registration. In particular, a direction estimate obtained from the same IMU used for preintegration is not an independent observation. We refer to this as a *same-IMU direction factor*. We test whether closer direction agreement lowers trajectory error, and whether the answer depends on factor weight or on whether gravity is estimated online.

Runtime correction nulling cannot test state removal. Suppressed coordinates remain in the covariance and influence retained variables through cross-correlation. We instead compile four FAST-LIO2 state configurations, each with a manifold that retains or removes gravity and accelerometer bias. The front end, data, initialization, calibration, and parameter settings remain fixed. A true 2×2 LIO-SAM state ablation provides a descriptive cross-architecture check. Within that graph, factor on/off and fixed/online gravity isolate the same-IMU direction factor.

Stress tests cover geometry degradation, periodic 1–5-s LiDAR absence, IMU weighting, initialization error, and dynamic startup. A history-matched switch separates fixing gravity from differences accumulated before an outage. We report vertical position RMSE (RMSE_z) and 3D position RMSE (ATE), since vertical improvement can accompany a worse trajectory. Absolute differences help judge whether a large percentage change is practically important.

This work contributes:

- within-system gravity and bias ablations in FAST-LIO2 and LIO-SAM, using actual state removal rather than suppressed corrections;
- tests of when state removal matters, including matched recovery after LiDAR outages and sensitivity to motion during initialization; and
- a direction-factor study showing that a smaller same-IMU residual does not guarantee lower vertical or 3D position error.

Default: keep both $\mathbf{g}$ and $\mathbf{b}_a$ online. The tested same-IMU direction factor is not a generic z -drift remedy.

II. RELATED WORK

State representation. Modern LIO includes iterated filters, smoothers, and continuous-time estimators. FAST-LIO/FAST-LIO2 place gravity on S^2 inside an iterated error-state filter [1], [5], [6], and Point-LIO retains the same convention [2]. VE-LIOM also estimates gravity online in an optimization framework [7], whereas LIO-SAM initializes a gravity-aligned frame without an equivalent online direction state [3]. Related pipelines also differ in feature construction and map access. LOAM introduced edge/plane registration [8], while FAST-LIO2 couples raw points to an incremental k -d tree [9]. Direct and continuous-time systems change scan matching and propagation together [10]–[12]. Cross-system accuracy therefore confounds gravity handling with the front end, trajectory model, map, and parameter settings.

Our primary intervention changes only the compiled state manifold; LIO-SAM provides a descriptive transfer check.

Gravity–attitude–bias observability. Gravity direction, roll/pitch, and accelerometer bias have coupled linearizations. Visual–inertial consistency analyses identify unobservable and weakly observable directions [13], [14], and planar motion further limits excitation [15]. LIO initializers consequently estimate gravity, biases, timing, and extrinsics before normal operation [16]; joint on-manifold calibration makes the coupling explicit [17]. These analyses establish when gravity is identifiable, but not whether its continued freedom improves pose under frequent LiDAR correction. We test the latter while intervening independently on $\mathbf{g}$ and $\mathbf{b}_a$. Mature visual–inertial systems combine observability decisions with initialization, relocalization, and map management [18]; their aggregate robustness cannot isolate one state’s post-initialization utility.

Gravity-enhanced residuals. Gravity-constrained registration removes rotational freedom using an IMU-derived vertical direction [19]. Recent radar–LiDAR and radar–leg estimators add velocity-supported or soft S^2 gravity information and report improved vertical accuracy [20], [21]. Elevator-specific models likewise show that non-inertial motion can violate a nominal gravity model [22]. These methods add sensor information or residual constraints; fixing an initialized gravity state does neither. We therefore cross a same-IMU gravity-direction factor with fixed/online gravity and pair its residual reduction with trajectory error. This design tests whether tighter down-direction consistency reliably reduces vertical drift.

Degradation and correction absence. Perceptual degeneracy is commonly detected from LiDAR geometry or estimator observability and handled through selective updates, alternate odometry, or direction-dependent weighting [23], [24]. Weak geometry and missing correction are often discussed within the same robustness setting, although their estimator inputs differ. We separate them experimentally: range/FoV degradation preserves the scan-update rhythm, whereas periodic dropout removes correction entirely. Because the expected effects are small, our protocol also audits reference-path plausibility and exact correction timestamps, consistent with uncertainty-aware benchmark generation [25]. Our claim concerns an observation regime, not a ranking of complete estimators. It distinguishes weak but accepted updates from rejected or unavailable ones, a difference hidden by a single trajectory score.

III. EXPERIMENTAL DESIGN

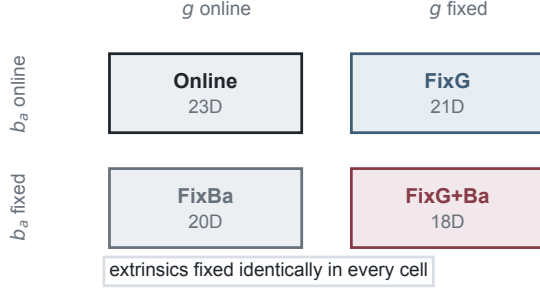
A. State Interventions and Controls

FAST-LIO2 is an iterated error-state filter on manifolds [1], [6]. Its state comprises position $\mathbf{p}$, orientation $\mathbf{R}$, LiDAR–IMU extrinsics $(\mathbf{R}_{LI}, \mathbf{t}_{LI})$, velocity $\mathbf{v}$, gyroscope bias $\mathbf{b}_g$, accelerometer bias $\mathbf{b}_a$, and fixed-magnitude gravity $\mathbf{g} \in S^2$:

$$d = 3\mathbf{p} + 3\mathbf{R} + 6_{\text{ext}} + 3\mathbf{v} + 3\mathbf{b}_g + 3\mathbf{b}_a + 2\mathbf{g} = 23. \quad (1)$$

Online extrinsic calibration is disabled in every configuration. The calibration remains necessary for deskewing and

a Two orthogonal state interventions



b FAST-LIO ROS1 RViz scan, three interventions

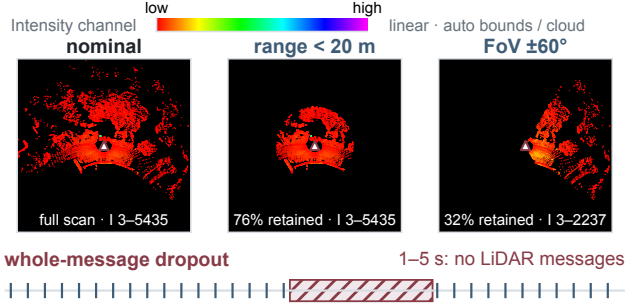


Fig. 2. Controlled design. **a**, Orthogonal compile-time state interventions; extrinsics are held fixed identically, not removed. **b**, One deskewed FAST-LIO scan, colored by retained point intensity under nominal, 20-m range, and $\pm 60^\circ$ FoV admission. The panels use the ROS1 RViz /cloud_registered intensity mapping with deterministic display decimation; dropout removes all LiDAR messages during 1–5-s gaps.

residual construction, but is not an experimental variable. The disputed coupling enters propagation as

$$\delta \dot{\mathbf{v}} \approx \delta \mathbf{g} - \mathbf{R} \delta \mathbf{b}_a - \mathbf{R} [\mathbf{a}_m - \mathbf{b}_a] \times \delta \boldsymbol{\theta}. \quad (2)$$

State dimension is fixed at compile time, so four executables implement the four configurations using otherwise identical code, front end, map, calibration, initialization, and parameter settings. ONLINE estimates $\mathbf{g}$ and $\mathbf{b}_a$ online (23D); FixG fixes $\mathbf{g}$ (21D); FixBa fixes $\mathbf{b}_a$ (20D); and FixG+Ba fixes both (18D). Fixed quantities are removed from the estimated state and retained as initialized constants in propagation. For comparison, a 23D proxy zeros their corrections without changing the manifold. Because this proxy retains their covariance blocks and influence on the Kalman gain, it tests the validity of correction nulling, not state removal.

We repeat the state intervention in LIO-SAM [3]. Native FG–BA fixes gravity and estimates $\mathbf{b}_a$; FG–B0 also removes $\mathbf{b}_a$ and its random walk. GE–BA/GE–B0 introduce a shared 2D, fixed-magnitude online gravity direction, while $\mathbf{b}_g$ remains online. The state dimension changes explicitly rather than through a near-zero prior. The full ablation spans 4 sequences and serves as a descriptive cross-architecture check, not a second equivalence population.

B. Correction Regimes and Direction Factor

The FAST-LIO2 nominal population contains 12 public sequences from MCD, TIERS, and M2DGR [26]–[28]. They span vehicle, handheld, and quadruped platforms with four

LiDAR/IMU combinations; path lengths range from 35 m to 3.2 km. We modify one MCD vehicle sequence offline by capping range at 20 m, restricting horizontal field of view (FoV) to a forward $\pm 60^\circ$ sector, or deleting consecutive LiDAR messages over 1-, 2-, 3-, or 5-s intervals repeated every 20 s. The geometric cuts remove distant returns or lateral/rear coverage while preserving scan timestamps. They are fixed geometric controls, not calibrated models of sensor failure. All four state configurations are run three times for the 2- and 3-s gaps; a TUHH handheld trajectory provides a separate test of 2/3/5-s gaps. Only the point-cloud stream changes. Paired configurations share the same IMU data, calibration, reference trajectory, first/last timestamps, and outage schedule. Here, *weak geometry* means accepted updates with fewer spatial constraints; *correction absence* means no LiDAR update during an interval (Fig. 2). The known dropout schedule isolates correction absence without modeling front-end rejection or update delays.

The IMU-weight sweep pairs ONLINE/FixG and scales IMU noise and bias random walk from 0.01 to 10 on 2 vehicle trajectories. A separate 12-run test injects $0.5^\circ/1^\circ/2^\circ$ gravity-direction errors after shared initialization on one vehicle and one handheld trajectory, then pairs ONLINE/FixG. The startup experiment compares all four configurations on 60-s suffixes. Each trajectory contributes one quasi-static and 3 low-rate phases, supplemented by 4 stronger phases selected without viewing pose outcomes. Within each phase, all configurations use the same bag offset. A 150-ms IMU/ground-truth descriptor selects phases before pose evaluation. The resulting 12 phases and 48 trajectories test the shared acceleration-mean initializer, not a motion-aware alternative.

A 10-run, 60-s allocation test on one vehicle and one handheld trajectory tilts $\mathbf{g}$ by $\pm 2^\circ$ alone or pairs it with $\mathbf{b}_{a,1} = \mathbf{b}_{a,0} + \mathbf{R}_0^\top (\mathbf{g}_1 - \mathbf{g}_0)$, preserving $\mathbf{g} - \mathbf{R}_0 \mathbf{b}_a$ initially. Both states remain online; the test probes allocation ambiguity, not absolute bias calibration.

A runtime audit alternates ONLINE/FixG over 3 serial pairs on MCD NTU Day10. After 100 warm-up scans, 3144 matched scans per run provide core, matching, and filter-algebra times; the last is cumulative update minus Jacobian construction. Whole-core timing remains descriptive because map workloads bifurcate.

To control for pre-outage history, we fix gravity at its Online mean at a pre-specified trigger and condition the retained covariance as $P_{x|g} = P_{xx} - P_{xg} P_{gg}^{-1} P_{gx}$, then continue propagation and measurement updates in a 21D error subspace. Unlike independently initialized FixG, this matched switch preserves the pre-switch state, map, and posterior history; only uncertainty changes at the trigger. Clean/dropout pairs share the trigger and are bit-identical before switching. Each original-phase pair is run three times serially, and two further pre-specified dropout starts test sensitivity to outage phase on each trajectory. For either error metric E , the interaction is the switch-minus-Online difference under dropout minus the same difference under clean input, reported in meters.

To test the added residual, the LIO-SAM graph receives at each post-initialization LiDAR correction epoch

$$\mathbf{r}_{d,k} = \mathbf{B}(\mathbf{d}_k^{\text{AHRS}})^\top \frac{\mathbf{R}_k^\top \hat{\mathbf{g}}}{\|\hat{\mathbf{g}}\|}, \quad (3)$$

where $\mathbf{d}_k^{\text{AHRS}}$ is the body-frame down direction from the latest attitude and heading reference system (AHRS) quaternion. The normalized $\mathbf{R}_k^\top \hat{\mathbf{g}}$ is the direction predicted from orientation and the fixed or estimated gravity direction. The columns of $\mathbf{B}$ form an orthonormal tangent basis at $\mathbf{d}_k^{\text{AHRS}}$. This is GTSAM’s two-component Unit3 direction residual, not an exact spherical logarithm. Reported direction-residual norms scale $\|\mathbf{r}_{d,k}\|$ by $180/\pi$ and are small-angle equivalents. AHRS estimation and preintegration use measurements from the same IMU, so the factor adds no independent sensor. It observes neither yaw, height, nor vertical velocity. We compare factor-off with $\sigma_d \in \{0.5^\circ, 2^\circ, 5^\circ\}$ and fixed/online gravity on two continuous-correction sequences. Under 5-s gaps, Hall05 and TUHH Day04 (handheld) each receive the full 2×3 gravity-state-by-weight design, with 3 serial factor-on/off pairs for each state–weight combination. We report each trajectory separately. Factor covariance is a controlled weight, not a claim that AHRS error is independent of preintegration. Factor effects use the state-and repeat-matched off run from this campaign, not the separate state-ablation baseline. The factor belongs to LIO-SAM’s incremental IMU–LiDAR estimation graph, not the downstream loop-closure pose graph. Gravity priors in global pose-graph optimization (PGO) are outside this intervention.

C. Inference and Admission

We report vertical position RMSE (RMSE_z) and 3D position RMSE (ATE) after rigid $SE(3)$ alignment, without scale fitting. FAST-LIO2 fits a prefix that spans both 10 s and 30 m; LIO-SAM uses only the time criterion. The protocol is fixed within each system and is not used to rank absolute errors across systems. Tables abbreviate RMSE_z as z . Percent change is used only within a paired trajectory and is interpreted alongside absolute error. The nominal FixG comparison uses two one-sided tests (TOST) on paired log ratios, with a pre-specified $\pm 5\%$ margin and 90% confidence intervals. We also report paired differences in meters because ratios magnify small changes around accurate baselines. A post-hoc $\pm 2\%$ TOST and equal-weight trajectory-family analysis are sensitivity checks, not replacement tests. A paired Wilcoxon test evaluates the factorial interaction $I = \Delta_{g,b_a} - \Delta_g - \Delta_{b_a}$. LIO-SAM and mechanism sweeps remain descriptive; no cross-trajectory pooled inference is made.

Admission requires six gates: plausible reference arc, matched attitude frame, exact modal frame count, no host sleep, launch-specific binary fingerprint, and serial execution. The early FAST-LIO2 logger hashed the main rather than each reduced executable; state logs and build times support the intended structures, but their exact hashes cannot be recovered. Later batches use launch-specific fingerprints. Structural audits require removed states to remain bit-exact,

retained states to move, $\|\mathbf{g}\| = 9.8090 \text{ m/s}^2$, and initial direction agreement within 0.1° . LIO-SAM additionally checks correction times, state dimension, factor count, AHRS age, and resets. A reset contributes a stability outcome but no accuracy value. Complete finite trajectories outside the estimated-arc diagnostic band remain as “A” warnings to avoid outcome-conditioned admission. A Hall05 run with misaligned correction timestamps was excluded and rerun; the final reset was not replaced.

For the matched APE traces, each pair shares the rigid transform fitted to its Online control; the plotted center and band are the median and full range of the three serial pairs. This preserves the zero pre-switch contrast and avoids selecting a representative failure run.

IV. RESULTS

A. Structural Validation

Structural logs verify the intended intervention before accuracy is examined. Online gravity moves by as much as 4.0° , while every removed gravity trace is bit-exact and retains the prescribed magnitude. Online $\mathbf{b}_a$ changes and fixed $\mathbf{b}_a$ does not. Paired correction timestamps, admitted frame counts, and initial directions match. The early fingerprint limitation is stated in Sec. III. These logs confirm distinct state treatments, not an accuracy benefit. The FAST-LIO2 nominal block contains 48 admitted trajectories across the four configurations. In LIO-SAM, exact state dimensions match the intended structures, and every admitted factor run contributes one direction factor at each audited epoch. The final Hall05 reset is retained as a stability outcome without an accuracy value; the earlier timestamp-misaligned run was excluded and rerun.

B. Continuous-Correction State Ablation

Across the 12 nominal sequences, the signed median RMSE_z change for FixG is $+0.9\%$, and the median absolute change is 1.1% (range -6.7% – $+2.6\%$). The corresponding ATE values are -0.1% and 0.9% (range -4.0% – $+5.4\%$). TOST places both mean paired effects inside the pre-specified $\pm 5\%$ margin (both p values $< 10^{-4}$). Back-transformed mean effects and 90% intervals are -0.11% $[-1.54, +1.34]\%$ for RMSE_z and $+0.09\%$ $[-1.16, +1.35]\%$ for ATE (Fig. 3; Table I). This supports average practical equivalence, not a per-trajectory guarantee. Both metrics also pass a post-hoc $\pm 2\%$ sensitivity test (maximum p 0.020), while equal weighting across five trajectory families retains the pre-specified decision (maximum p 0.002). A launch-fingerprint reproduction on the same sequence units gives -0.28% $[-1.84, +1.30]\%$ – -0.11% $[-1.50, +1.31]\%$ and passes both margins (maximum strict p 0.040); it is a reproducibility check, not an independent $n = 24$ population. The paired trace in Fig. 3b shows a moving gravity state alongside nearly superposed vertical estimates. The equivalence result comes from the paired population test, not this individual trace.

Absolute scale matters: median absolute FixG–Online differences are 0.017 m for RMSE_z and 0.014 m for ATE. On m2dgr-hall, $+2.3\%$ is only $+1.0 \text{ mm}$ in RMSE_z ; its

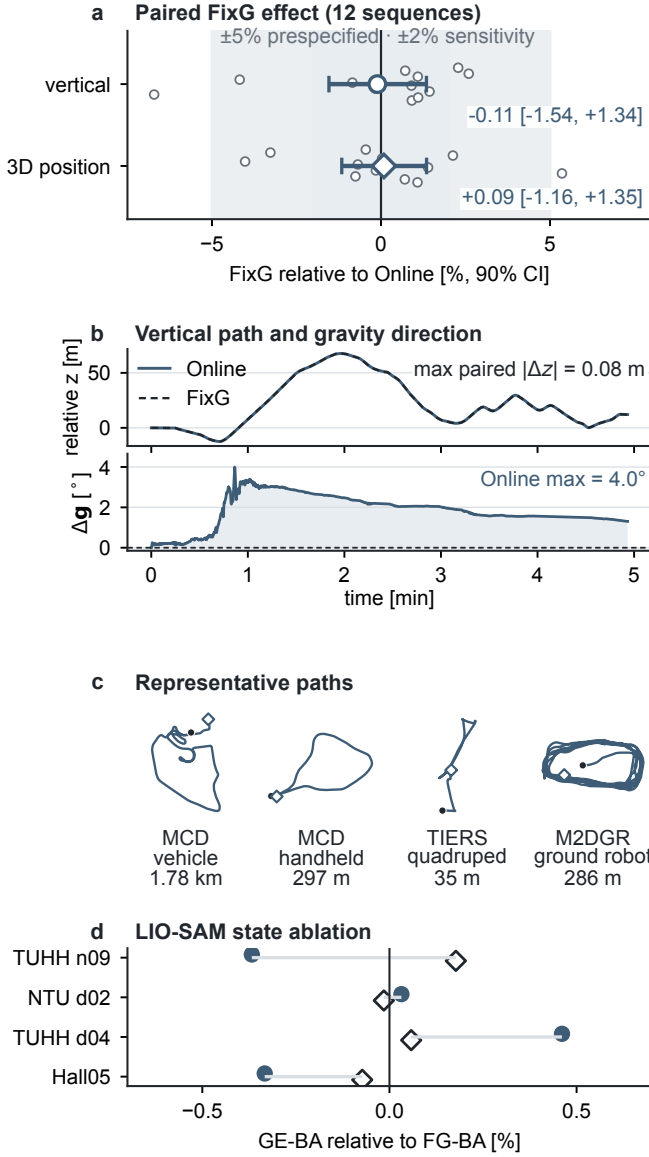


Fig. 3. Continuous-correction state ablation. **a**, FAST-LIO2 mean paired effects with 90% confidence intervals ($n = 12$ paired sequences; small symbols show individual effects). **b**, A paired sequence in which online gravity moves while Online and FixG vertical estimates remain nearly superposed; this trace is illustrative, not the population test. **c**, Representative ONLINE trajectories; XY scales are independent, and path lengths come from admitted references. **d**, Descriptive LIO-SAM transfer; circles denote RMSE_z and diamonds ATE (3D position error).

ATE change is -0.45% (-5.7 mm). We therefore interpret paired percentages with absolute differences and require both metrics, which can disagree in sign.

The LIO-SAM state ablation yields a similar nominal pattern. Across 4 complete sequences, GE-BA relative to native FG-BA changes RMSE_z by -0.37/+0.46% and ATE by -0.07/+0.18%; neither metric has a consistent sign. Fixing both quantities changes RMSE_z/ATE by -3.20/+0.76%/-1.09/+0.27%. These data provide a descriptive cross-architecture check on the absent nominal gain from online gravity. They do not establish LIO-SAM equivalence or a general rule for **b_a**.

TABLE I
STATE ABLATION WITH CONTINUOUS CORRECTION.

FAST-LIO2 paired change vs. ONLINE [%], $n = 12$				
Configuration	dim.	median $z/\Delta TE$	range z	range ATE
FixG	21	+0.9 / -0.1	[-6.7, +2.6]	[-4.0, +5.4]
FixBA	20	+1.5 / +0.2	[-2.1, +21.9]	[-3.3, +12.6]
FixG+BA	18	+1.9 / +0.2	[-8.7, +47.2]	[-6.1, +14.6]
LIO-SAM paired $\Delta z/\Delta ATE$ vs. FG-BA [%]				
Sequence		FG-B0	GE-BA	GE-B0
Hall05		+0.76 / +0.02	-0.33 / -0.07	+1.27 / +0.01
TUHH d04		-3.20 / -1.09	+0.46 / +0.06	+0.93 / +0.16
NTU d02		+0.04 / +0.02	+0.03 / -0.01	+0.00 / -0.04
TUHH n09		-0.22 / +0.27	-0.37 / +0.18	-0.09 / +0.46

C. Weak Geometry and Correction Absence

Continuously degraded scans do not reproduce a dropout penalty. A 20m range cap changes FixG RMSE_z/ATE by +0.2%/-4.0%; a forward $\pm 60^\circ$ FoV changes them by +6.9%/+3.5%. Both retain scan-update timing (Table II).

Complete dropout separates the designs. A 1-s gap every 20s remains benign. At 2s, vehicle RMSE_z/ATE move in opposite directions (-11.2%/+16.8%), so neither metric alone supports a configuration choice. At 3s, both rise by +93.1%/+119.9% in all 3 serial repeats. The handheld response is later: +8.0%/+7.3% at 3s and +27.9%/+34.5% at 5s. Vehicle 5-s effects remain worse than Online but are smaller than at 3s, so duration is not a monotonic dose.

The matched APE traces in Fig. 4c,d start at zero, stay near zero with continuous correction, but separate progressively after repeated gaps. The vehicle contrast initially changes sign; the handheld responds later. The repeated trajectories do not show an effect confined to the first return. The 2–3-s vehicle transition and later handheld response place the onset at different gap durations, rather than a universal threshold.

The matched switch reproduces the regime contrast across 3 serial pairs at the original dropout phase. Clean-input ATE effects are small: -0.83% on the vehicle and -0.30% on the handheld trajectory. The median dropout-minus-clean ATE interactions are +4.16m and +1.01m, respectively. Across 3 dropout phases, ATE interactions are positive on both trajectories, whereas RMSE_z interactions cross zero (Table II). The stable effect across these phases is therefore in 3D position error, not vertical error.

D. Direction-Factor Response

Nominally, factor strength has no monotonic dose response (Fig. 5; Table III). With fixed gravity, the tightest factor worsens RMSE_z by +1.0% to +1.2%, whereas the intermediate factor changes RMSE_z/ATE by -1.7% to -1.2%/-0.75% to +0.03%. Online-gravity ATE stays within -0.07% to +0.04%. Across these configurations, smaller direction residuals do not predict smaller trajectory errors.

Under repeated 5-s gaps, the two trajectories respond differently. On Hall05, all 6 state-weight medians improve both errors, but the FixG/5° configuration yields only 2/3 accuracy runs because one planned run resets; that failure is counted and was not replaced. TUHH contains both sign

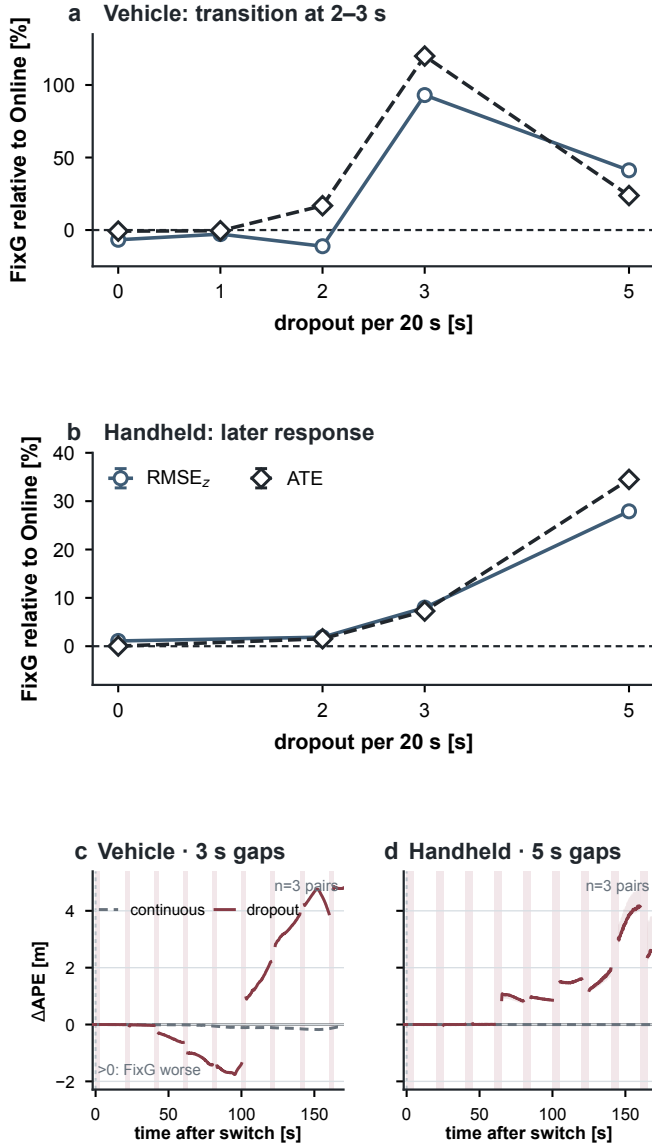


Fig. 4. Correction-absence boundary. **a,b**, FixG relative to ONLINE on vehicle and handheld trajectories; both RMSE_z and ATE are required. **c,d**, APE difference after the matched 23D-to-21D switch: median and range across 3 serial pairs under a common alignment. Shading denotes missing corrections; positive values favor Online.

reversal and metric disagreement. With FixG/0.5°, RMSE_z changes by -64.6% while ATE changes by +97.6%. With online gravity/2°, both deteriorate by +450.0%/+108.4%. The 2 complete TUHH trajectories outside the estimator arc diagnostic band retain their metrics and carry “A” warnings; neither was replaced. Thus a lower vertical error can conceal worse 3D pose, and a direction factor can worsen both metrics. Neither increasing factor strength nor keeping gravity online ensures an improvement.

E. Accelerometer Bias and Gravity-Bias Coupling

Accelerometer bias does not inherit gravity’s removal result. FixBA and FixG+BA have median RMSE_z changes of +1.5% and +1.9%, with larger sequence-level ranges than FixG (Table I). True-manifold interaction is -0.8 pp

TABLE II
CORRECTION-REGIME BOUNDARY.

Condition	ONLINE z /ATE [m]	Δz [%]	Δ ATE [%]	n
VEHICLE continuous correction				
nominal	2.49/6.60	-6.7	-0.8	1
range 20 m	2.81/10.08	+0.2	-4.0	1
FoV $\pm 60^\circ$	5.17/6.49	+6.9	+3.5	1
VEHICLE dropout per 20 s				
1 s/20 s	2.43/6.50	-2.8	-0.5	1
2 s/20 s	2.31/6.27	-11.2	+16.8	3
3 s/20 s	3.09/8.42	+93.1	+119.9	3
5 s/20 s	4.92/13.65	+41.2	+23.8	1
HANDHELD dropout per 20 s				
2 s/20 s	0.445/0.519	+1.9	+1.5	1
3 s/20 s	0.485/0.536	+8.0	+7.3	1
5 s/20 s	3.28/4.07	+27.9	+34.5	1
MATCHED POSTERIOR SWITCH dropout-clean interaction [m]				
vehicle repeats	-	+0.62	+4.16	3
vehicle phases	-	[-0.23, +2.74]	[+0.87, +4.16]	3
handheld repeats	-	+0.09	+1.01	3
handheld phases	-	[-0.08, +0.22]	[+0.30, +1.09]	3

($p = 0.470$) for RMSE_z and -0.8 pp ($p = 0.151$) for ATE, providing no evidence of superadditivity. The correction-nulling proxy biases the fixed- $\mathbf{b}_a$ comparison by +7.3 pp, confirming that it is not true removal.

The paired intervention shifts $\mathbf{b}_a$ by 0.342 m/s² while preserving $\mathbf{g} - \mathbf{R}\mathbf{b}_a$ within 2.0×10^{-7} m/s². In the handheld final 10 s, gravity/bias remain 0.73–0.74°/0.122–0.126 m/s² apart, but the effective term/position differ by only 0.0077–0.0091 m/s²/4.2–8.4 mm; the vehicle estimates return close to baseline. Handheld poses thus remain similar despite distinct gravity and bias estimates.

F. IMU Weight and Initialization

IMU weighting does not reveal a uniform advantage for either gravity choice. On vehicle NTU Day10, FixG changes RMSE_z by -10.7% to -5.0% and ATE by -1.2% to -0.2%. On vehicle NTU Night04, RMSE_z changes by -3.2% to +1.7%, but ATE worsens by +2.3% to +9.0%. Maximum roll/pitch change is 0.052°.

The initialization-direction stress test also shows no monotonic FixG penalty. Across 12 paired runs at 0.5°/1°/2°, RMSE_z/ATE changes span -3.5% to +2.1%/-2.3% to +1.8%, and the maximum roll/pitch change is 0.013°.

Across 12 start phases and 48 trajectories, the dynamic-start 2×2 yields no universal gravity choice. At one stronger vehicle phase, Online and FixG produce 0.905 m and 5.074 m ATE, respectively, while RMSE_z reverses (0.625 m versus 0.170 m). Fixed bias is more sensitive to startup phase: both fixed-bias configurations diverge at one low-rate handheld start. FixBa also worsens ATE in all 4 stronger-motion phases (+1.64% to +65.82%), although it helps at other low-rate phases. The 2 complete, finite failure outcomes are retained rather than rerun. The phase- and metric-dependent signs do not yield a uniform vertical Online gain.

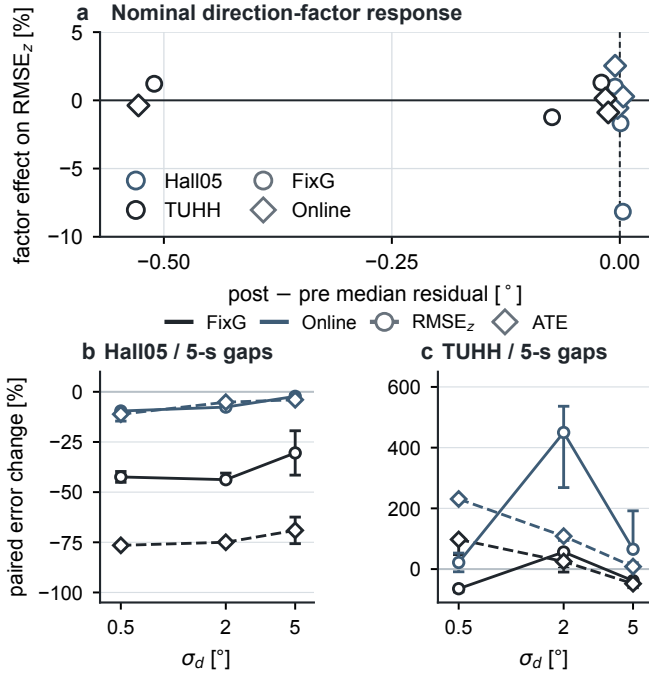


Fig. 5. Direction-factor response. **a**, Nominal residual-norm (degree-equivalent) and RMSE_z changes; color denotes trajectory, shape denotes gravity state. **b,c**, Dropout effects: paired medians and full ranges, with lines connecting tested weights. Positive means worse. Failures and warnings remain in Table III.

V. DISCUSSION

A. Why Scan Recovery Changes the Comparison

Accepted LiDAR scans repeatedly constrain pose, so gravity-state drift need not produce a worse trajectory. During an outage, errors accumulate until registration resumes from the propagated state.

For a small gravity-direction discrepancy ε propagated without an external update for time T , the leading position contribution is

$$\|\delta \mathbf{p}_g\| \approx \frac{1}{2} g \sin(\varepsilon) T^2. \quad (4)$$

This describes displacement before recovery, not final trajectory error. Registration may correct a large error but leave a smaller one partly uncorrected, depending on the scene and returning pose. This is consistent with the non-monotonic 3-s and 5-s results, although we do not directly measure the registration convergence basin.

In the matched switch, both runs start from the same state and map, but fixing gravity conditions the covariance. Gravity has no process dynamics, so the nominal means remain equal during the first gap; Eq. 4 cannot explain separation before scan return. Differences arise during correction, through the changed covariance and continued Online gravity updates. The APE traces show accumulation over successive recovery corrections. ATE interactions stay positive across tested starts, while RMSE_z interactions change sign. The repeated advantage concerns 3D recovery, not direct height information.

The relevant interval $T_k = t_k - t_{k-1}$ is between *accepted pose corrections*, not arriving LiDAR packets. A rejected

TABLE III
DIRECTION-FACTOR EFFECTS RELATIVE TO FACTOR-OFF.

Weight	g state	ΔRMSE_z [%]	ΔATE [%]	outcome
Hall05 5-s dropout per 20 s				
0.5°	FixG	-42.4	-76.5	3/3
2°	FixG	-43.8	-75.0	3/3
5°	FixG	-30.5	-69.1	2/3; 1 R
0.5°	Online	-9.6	-11.2	3/3
2°	Online	-7.6	-5.2	3/3
5°	Online	-2.3	-4.0	3/3
TUHH HANDHELD 5-s dropout per 20 s				
0.5°	FixG	-64.6	+97.6	3/3; 1 A
2°	FixG	+56.2	+25.0	3/3
5°	FixG	-38.3	-48.1	3/3
0.5°	Online	+21.8	+230.9	3/3; 1 A
2°	Online	+450.0	+108.4	3/3
5°	Online	+65.4	+8.2	3/3

Outcome entries report accuracy runs/planned runs; R denotes a graph reset and A a retained arc warning.

scan cannot interrupt inertial error accumulation. Message removal controls these intervals, whereas natural rejection may depend on geometry and the estimated pose. Biased but accepted registrations instead introduce erroneous corrections, a different failure mode not tested here. Scan rate alone is therefore insufficient to transfer the observed thresholds.

B. Interpreting the Direction-Factor Response

The factor uses AHRS directions derived from the IMU measurements also used for preintegration. It observes neither height nor vertical velocity, so its vertical-error effects are mediated by attitude, bias, propagation, and registration.

At fixed gravity magnitude, a direction error ε gives

$$\begin{aligned} \|\mathbf{e}_g^\top \delta \mathbf{g}\| &= g(1 - \cos \varepsilon) = O(\varepsilon^2), \\ \|\mathbf{P}_\perp \delta \mathbf{g}\| &= g \sin \varepsilon = O(\varepsilon). \end{aligned} \quad (5)$$

where $\mathbf{e}_g$ is unit true down and $\mathbf{P}_\perp = \mathbf{I} - \mathbf{e}_g \mathbf{e}_g^\top$. Small direction errors therefore act to first order transversely, but only to second order along true down. A direction factor can still change RMSE_z by rotating measured specific force or changing scan-matching initialization. Neither route measures height, and Eq. 5 alone does not determine the final error change.

Hall05's improvements and TUHH's mixed or adverse responses reject an unconditional benefit from stronger direction consistency. They do not invalidate independent gravity, velocity, contact, radar, or registration information [19]–[21], nor do they test a downstream global pose-graph prior. Factor covariance must be validated with the state structure, correction regime, and both trajectory metrics.

Unmodeled same-IMU correlation could contribute to the adverse response, but we do not isolate it from weighting and online-gravity interactions. A smaller residual establishes agreement with AHRS, not an independent reference (Fig. 5). Table III also retains resets and arc warnings; reporting only successful accuracy comparisons would overstate reliability.

C. Choosing a Default Configuration

Keeping $\mathbf{g}$ and $\mathbf{b}_a$ online allows adjustment but does not guarantee physical accuracy. The handheld runs retain

different gravity and bias estimates despite similar effective accelerations and poses. Vehicle excitation largely removes those differences. As Eq. 2 suggests, accurate pose does not establish that gravity and bias have each been identified.

With reliable initialization and continuous correction, FAST-LIO2 FixG is equivalent on average but saves negligible compute. Keeping both states online allows adaptation when motion contaminates startup or corrections disappear. Fixed-bias variants are phase-sensitive, and one FixG start incurs large ATE harm. We therefore recommend keeping both states online by default. FixG is an option for a validated steady-state setting, not a generally better estimator. A separate direction factor needs its own paired RMSE_z and ATE validation under the intended operating conditions.

Across 3 serial pairs, FixG reduced filter algebra by 7.5%, but it occupied only 0.83% of core time versus 88.2% for scan matching. Its attributable saving was 0.06%, and whole-core change (-11.6% to +3.1%) crossed zero. Dimension reduction is not a runtime strategy.

Two dropout and two initialization trajectories cannot establish universal thresholds. The factor study retains 1 reset and 2 arc warnings. Larger initialization errors, calibration faults, and motion-aware initialization also remain untested. TOST concerns pose error, not covariance consistency; dropout results are not pooled across trajectories.

VI. CONCLUSION

With continuous LiDAR correction, FAST-LIO2's mean paired effects of fixing gravity on vertical and 3D position error satisfy practical equivalence across 12 dataset sequences. A descriptive 4-sequence LIO-SAM ablation also finds no consistent benefit from online gravity estimation. This does not justify fixing gravity or bias by default. The matched-dropout and dynamic-start tests show recovery and failure costs that nominal averages miss, while the runtime saving is negligible. We recommend keeping both $\mathbf{g}$ and $\mathbf{b}_a$ online, without treating their individual estimates as calibrated measurements. FixG remains an optional, validated steady-state simplification. Do not add the tested same-IMU gravity-direction factor as a generic z -drift remedy; the factor can reduce its residual while improving, leaving unchanged, or worsening trajectory error. Independent gravity sensing and global loop-closure PGO priors remain outside this claim.

REFERENCES

- [1] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, "FAST-LIO2: Fast direct LiDAR-inertial odometry," *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2053–2073, 2022.
- [2] D. He, W. Xu, N. Chen, F. Kong, C. Yuan, and F. Zhang, "Point-LIO: Robust high-bandwidth LiDAR-inertial odometry," *Advanced Intelligent Systems*, vol. 5, no. 7, p. 2200459, 2023.
- [3] T. Shan, B. Englot, D. Meyers, W. Wang, C. Ratti, and D. Rus, "LIO-SAM: Tightly-coupled LiDAR inertial odometry via smoothing and mapping," in *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2020, pp. 5135–5142.
- [4] X. Gao, "Lightning-LM," GitHub repository, 2026, commit 1325fed. [Online]. Available: <https://github.com/gaoxiang12/lightning-lm>
- [5] W. Xu and F. Zhang, "FAST-LIO: A fast, robust LiDAR-inertial odometry package by tightly-coupled iterated Kalman filter," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 3317–3324, 2021.
- [6] D. He, W. Xu, and F. Zhang, "Kalman filters on differentiable manifolds," *arXiv preprint arXiv:2102.03804*, 2021.
- [7] Y. Gao and L. Zhao, "VE-LIOM: A versatile and efficient LiDAR-inertial odometry and mapping system," *Remote Sensing*, vol. 16, no. 15, p. 2772, 2024.
- [8] J. Zhang and S. Singh, "LOAM: Lidar odometry and mapping in real-time," in *Robotics: Science and Systems (RSS)*, 2014.
- [9] Y. Cai, W. Xu, and F. Zhang, "ikd-tree: An incremental KD tree for robotic applications," *arXiv preprint arXiv:2102.10808*, 2021.
- [10] Z. Wang, L. Zhang, Y. Shen, and Y. Zhou, "D-LIOM: Tightly-coupled direct LiDAR-inertial odometry and mapping," *IEEE Transactions on Multimedia*, vol. 25, pp. 3905–3920, 2023.
- [11] T.-M. Nguyen, D. Duberg, P. Jensfelt, S. Yuan, and L. Xie, "SLICT: Multi-input multi-scale surfel-based LiDAR-inertial continuous-time odometry and mapping," *IEEE Robotics and Automation Letters*, vol. 8, no. 4, pp. 2102–2109, 2023.
- [12] K. Chen, R. Nemiroff, and B. T. Lopez, "Direct LiDAR-inertial odometry: Lightweight LIO with continuous-time motion correction," in *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2023, pp. 3983–3989.
- [13] J. A. Hesch, D. G. Kottas, S. L. Bowman, and S. I. Roumeliotis, "Consistency analysis and improvement of vision-aided inertial navigation," *IEEE Transactions on Robotics*, vol. 30, no. 1, pp. 158–176, 2014.
- [14] T. Qin, P. Li, and S. Shen, "VINS-Mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Transactions on Robotics*, vol. 34, no. 4, pp. 1004–1020, 2018.
- [15] K. J. Wu, C. X. Guo, G. Georgiou, and S. I. Roumeliotis, "VINS on wheels," in *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2017, pp. 5155–5162.
- [16] F. Zhu, Y. Ren, and F. Zhang, "Robust real-time LiDAR-inertial initialization," in *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2022, pp. 3948–3955.
- [17] R. Nemiroff, K. Chen, B. Morrell, A.-a. Agha-mohammadi *et al.*, "Joint on-manifold gravity and accelerometer intrinsics estimation for inertially aligned mapping," *arXiv preprint arXiv:2303.03505*, 2023.
- [18] C. Campos, R. Elvira, J. J. G. Rodríguez, J. M. M. Montiel, and J. D. Tardós, "ORB-SLAM3: An accurate open-source library for visual, visual-inertial, and multimap SLAM," *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.
- [19] V. Kubelka, M. Vaidis, and F. Pomerleau, "Gravity-constrained point cloud registration," in *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2022, pp. 4873–4879.
- [20] C. Noh, W. Yang, M. Jung, S. Jung, and A. Kim, "GaRLIO: Gravity enhanced radar-LiDAR-inertial odometry," *arXiv preprint arXiv:2502.07703*, 2025.
- [21] C. Noh, S. Jung, H. Kim, Y. Hu, L. Herlant, and A. Kim, "GaRLILEO: Gravity-aligned radar-leg-inertial enhanced odometry," *arXiv preprint arXiv:2511.13216*, 2025.
- [22] Y. Zhang, Y. Huang, Y. Zhang, C. Li, H. Liu, and M. Yang, "Elevator-LIO: Robust LiDAR-inertial odometry for multi-floor navigation under elevator-induced non-inertial motion," *arXiv preprint arXiv:2605.24495*, 2026.
- [23] A. Tagliabue, J. Tordesillas, X. Cai, A. Santamaria-Navarro, J. P. How, L. Carlone, and A.-a. Agha-mohammadi, "LION: Lidar-inertial observability-aware navigator for vision-denied environments," *arXiv preprint arXiv:2102.03443*, 2021.
- [24] G. Yao, H. Wang, and Q. Chang, "D²-LIO: Enhanced optimization for LiDAR-IMU odometry considering directional degeneracy," *arXiv preprint arXiv:2508.14355*, 2025.
- [25] X. Hu, L. Zheng, J. Wu, R. Geng, Y. Yu, H. Wei, X. Tang, L. Wang, J. Jiao, and M. Liu, "PALoc: Advancing SLAM benchmarking with prior-assisted 6-DoF trajectory generation and uncertainty estimation," *IEEE/ASME Transactions on Mechatronics*, vol. 29, no. 6, pp. 4297–4308, 2024.
- [26] T.-M. Nguyen, S. Yuan, T. H. Nguyen, P. Yin, H. Cao, L. Xie, M. Wozniak, P. Jensfelt, M. Thiel, J. Ziegenbein, and N. Blunder, "MCD: Diverse large-scale multi-campus dataset for robot perception," in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 22 304–22 313.
- [27] H. Sier, Q. Li, X. Yu, J. Peña Queraltá, Z. Zou, and T. Westerlund, "A benchmark for multi-modal LiDAR SLAM with ground truth in GNSS-denied environments," *Remote Sensing*, vol. 15, no. 13, p. 3314, 2023.
- [28] J. Yin, A. Li, T. Li, W. Yu, and D. Zou, "M2DGR: A multi-sensor and multi-scenario SLAM dataset for ground robots," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 2266–2273, 2022.